

CHAPTER 2

SIMPLE LINEAR REGRESSION

2.1 SIMPLE LINEAR REGRESSION MODEL

This chapter considers the **simple linear regression model**, that is, a model with a single regressor x that has a relationship with a response y that is a straight line. This simple linear regression model is

$$y = \beta_0 + \beta_1 x + \varepsilon \quad (2.1)$$

where the intercept β_0 and the slope β_1 are unknown constants and ε is a random error component. The errors are assumed to have mean zero and unknown variance σ^2 . Additionally we usually assume that the errors are uncorrelated. This means that the value of one error does not depend on the value of any other error.

It is convenient to view the regressor x as controlled by the data analyst and measured with negligible error, while the response y is a random variable. That is, there is a probability distribution for y at each possible value for x . The mean of this distribution is

$$E(y|x) = \beta_0 + \beta_1 x \quad (2.2a)$$

and the variance is

$$\text{Var}(y|x) = \text{Var}(\beta_0 + \beta_1 x + \varepsilon) = \sigma^2 \quad (2.2b)$$

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Thus, the mean of y is a linear function of x although the variance of y does not depend on the value of x . Furthermore, because the errors are uncorrelated, the responses are also uncorrelated.

The parameters β_0 and β_1 are usually called **regression coefficients**. These coefficients have a simple and often useful interpretation. The slope β_1 is the change in the mean of the distribution of y produced by a unit change in x . If the range of data on x includes $x = 0$, then the intercept β_0 is the mean of the distribution of the response y when $x = 0$. If the range of x does not include zero, then β_0 has no practical interpretation.

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

The parameters β_0 and β_1 are unknown and must be estimated using sample data. Suppose that we have n pairs of data, say $(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)$. As noted in Chapter 1, these data may result either from a controlled experiment designed specifically to collect the data, from an observational study, or from existing historical records (a retrospective study).

2.2.1 Estimation of β_0 and β_1

The **method of least squares** is used to estimate β_0 and β_1 . That is, we estimate β_0 and β_1 so that the sum of the squares of the differences between the observations y_i and the straight line is a minimum. From Eq. (2.1) we may write

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (2.3)$$

Equation (2.1) maybe viewed as a **population regression model** while Eq. (2.3) is a **sample regression model**, written in terms of the n pairs of data (y_i, x_i) ($i = 1, 2, \dots, n$). Thus, the least-squares criterion is

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad (2.4)$$

The least-squares estimators of β_0 and β_1 , say $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

and

$$\left. \frac{\partial S}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

Simplifying these two equations yields

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i x_i \end{aligned} \quad (2.5)$$

Equations (2.5) are called the **least-squares normal equations**. The solution to the normal equations is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (2.6)$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} \quad (2.7)$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

are the averages of y_i and x_i , respectively. Therefore, $\hat{\beta}_0$ and $\hat{\beta}_1$ in Eqs. (2.6) and (2.7) are the **least-squares estimators** of the intercept and slope, respectively. The fitted simple linear regression model is then

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \quad (2.8)$$

Equation (2.8) gives a point estimate of the mean of y for a particular x .

Since the denominator of Eq. (2.7) is the corrected sum of squares of the x_i and the numerator is the corrected sum of cross products of x_i and y_i , we may write these quantities in a more compact notation as

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} = \sum_{i=1}^n (x_i - \bar{x})^2 \quad (2.9)$$

and

$$S_{xy} = \sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n} = \sum_{i=1}^n y_i (x_i - \bar{x}) \quad (2.10)$$

Thus, a convenient way to write Eq. (2.7) is

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad (2.11)$$

The difference between the observed value y_i and the corresponding fitted value $\hat{y}_i$ is a **residual**. Mathematically the i th residual is

$$e_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i), \quad i = 1, 2, \dots, n \quad (2.12)$$

Residuals play an important role in investigating **model adequacy** and in detecting departures from the underlying assumptions. This topic is discussed in subsequent chapters.

Example 2.1 The Rocket Propellant Data

A rocket motor is manufactured by bonding an igniter propellant and a sustainer propellant together inside a metal housing. The shear strength of the bond between the two types of propellant is an important quality characteristic. It is suspected that shear strength is related to the age in weeks of the batch of sustainer propellant. Twenty observations on shear strength and the age of the corresponding batch of propellant have been collected and are shown in Table 2.1. The scatter diagram, shown in Figure 2.1, suggests that there is a strong statistical relationship between shear strength and propellant age, and the tentative assumption of the straight-line model $y = \beta_0 + \beta_1 x + \varepsilon$ appears to be reasonable.

TABLE 2.1 Data for Example 2.1

Observation, i	Shear Strength, y_i (psi)	Age of Propellant, x_i (weeks)
1	2158.70	15.50
2	1678.15	23.75
3	2316.00	8.00
4	2061.30	17.00
5	2207.50	5.50
6	1708.30	19.00
7	1784.70	24.00
8	2575.00	2.50
9	2357.90	7.50
10	2256.70	11.00
11	2165.20	13.00
12	2399.55	3.75
13	1779.80	25.00
14	2336.75	9.75
15	1765.30	22.00
16	2053.50	18.00
17	2414.40	6.00
18	2200.50	12.50
19	2654.20	2.00
20	1753.70	21.50